

Formula sheet

Price Return $PR_i = \frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}}$ $PR_I = \sum_{i=1}^n \omega_i PR_i = \sum_{i=1}^n \omega_i \left(\frac{P_{it} - P_{i(t-1)}}{P_{i(t-1)}} \right)$ $w_i = \text{weight of security}$

Total Return $TR_i = \frac{P_i - P_{i(t-1)} + \text{Div}}{P_{i(t-1)}}$ $w_i^P = \frac{P_i}{\sum_{i=1}^N P_i}$ $r^P = \sum_{i=1}^N w_i^P r_i$ $\text{Index}^P = \frac{\sum_{i=1}^N P_i}{d}$ $PW1$ $\hookrightarrow \text{no. in index}$

$PW1$ $\text{new divisor} = \frac{\text{sum of constituents before split}}{\text{index value before split}}$

MCWI $w_i^M = \frac{Q_i P_i}{\sum_{i=1}^N Q_i P_i}$ $\omega_i^{\text{float}} = \frac{f_i Q_i P_i}{\sum_{j=1}^N f_j P_j Q_j}$ $r^M = \sum_{i=1}^N w_i^M r_i$

MCWI $\text{Index}^M = \frac{\sum_{i=1}^N Q_i P_i}{d}$ f - shares outstanding in mkt float
 Q - no. of outstanding shares

MCWI $d = \frac{\sum_{i=1}^N Q_{Bi} P_{Bi}}{\text{Index}_B^M}$ EWI $w_i^E = \frac{1}{N}$ $r^E = \frac{\sum_{i=1}^N r_i}{N}$ $\text{Index}^E = \frac{\sum_{i=1}^N Q_i P_i}{d}$

EWI $d = \frac{\sum_{i=1}^N Q_{Bi}^* P_{Bi}}{\text{Index}_B^E}$ FWS $\omega_i = \frac{F_i}{\sum_{j=1}^n F_j}$ F_i - fundamental size feature of company

Utility $E(U) = \sum_w U(W)P(W)$ $ARA = \frac{-U''(W)}{U'(W)}$ $RRA = W \frac{-U''(W)}{U'(W)}$

$RP = \frac{1}{2} A(w) \cdot \sigma^2$ $\text{Variance} = \sigma^2 = \sum_{i=1}^n [R_i - E(R_i)]^2 P_i$

$\text{Standard Deviation} = \sigma = \sqrt{\sum_{i=1}^n [R_i - E(R_i)]^2 P_i}$ $\text{Cov}_{ij} = E \{ [R_i - E(R_i)][R_j - E(R_j)] \}$

$\rho_{i,j} = \frac{\sigma_{i,j}}{\sigma_i \sigma_j}$ $\sigma_p^2 = \omega_1^2 \sigma_1^2 + 2\omega_1 \omega_2 \rho_{12} \sigma_1 \sigma_2 + \omega_2^2 \sigma_2^2$ $U = E(r) - \frac{1}{2} A \sigma^2$

$\sigma_p^2 = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + \omega_3^2 \sigma_3^2 + 2\omega_1 \omega_2 \sigma_{1,2} + 2\omega_1 \omega_3 \sigma_{1,3} + 2\omega_2 \omega_3 \sigma_{2,3}$

$s = \frac{E(r_p) - r_f}{\sigma_p}$ $w_1 = \frac{R_1 \sigma_2^2 - R_2 \sigma_{1,2}}{R_1 \sigma_2^2 + R_2 \sigma_1^2 - (R_1 + R_2) \sigma_{1,2}}$ $y = \frac{E(r_p) - r_f}{A \sigma_p^2}$

$\omega_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$

$\omega_1 = \frac{\sigma_2^2 - \rho_{12} \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho_{12} \sigma_1 \sigma_2}$ $r_i = r_f + \beta(r_m - r_f), \text{ where } \beta_i = \frac{\sigma_{im}}{\sigma_m^2}$

$R_i = \alpha_i + \beta_i R_M + e_i$ $\sigma_i^2 = \beta_i^2 \sigma_m^2 + \sigma^2(e_i)$ $\text{Cov}(r_i, r_j) = \text{Cov}(\beta_i m + e_i, \beta_j m + e_j) = \beta_i \beta_j \sigma_m^2$

$$R_{it} = a_i + \sum_{j=1}^k b_{ij} F_{jt} + \varepsilon_{it} \quad ER_{it} = \lambda_0 + \sum_{j=1}^k b_{ij} \lambda_j$$

$$E(R_i) = \alpha_i + \beta_{iM}E(R_M) + \beta_{iSMB}E(R_{SMB}) + \beta_{iHML}E(R_{HMLt}) + e_{it}$$

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$$E(R_i) = \alpha_i + \beta_{iM}E(R_M) + \beta_{iSMB}E(R_{SMB}) + \beta_{iHML}E(R_{HMLt}) + \beta_{iCMA}E(R_{CMAt}) + \beta_{iRMW}E(R_{RMWt}) + e_{it}$$

$$\Delta_t = \sum_{i=1}^N w_i R_{it} - R_{bt} = R_{pt} - R_{bt} \quad \sigma_{\Delta}^2 = \frac{\sum_{t=1}^T (\Delta_t - \bar{\Delta})^2}{(T-1)} \quad \sigma_{\Delta} = \sqrt{\sigma_{\Delta}^2} = \text{periodic tracking error} \quad TE = \sigma_{\Delta} \sqrt{P}$$

$$w_i^0 = \frac{\alpha_i}{\sigma^2(e_i)} \quad w_i = \frac{w_i^0}{\sum_{i=1}^n w_i^0} \quad \alpha_A = \sum_{i=1}^n w_i \alpha_i \quad \sigma^2(e_A) = \sum_{i=1}^n w_i^2 \sigma^2(e_i)$$

$$w_A^0 = \left[\frac{\alpha_A / \sigma^2(e_A)}{RP_M / \sigma_M^2} \right] \quad \beta_A = \sum_{i=1}^n w_i \beta_i \quad w_A^* = \frac{w_A^0}{1 + (1 - \beta_A) w_A^0} \quad E(R_p) = (w_M^* + w_A^* \beta_A) RP_M + w_A^* \alpha_A$$

$$\sigma_p^2 = \beta P^2 \sigma_m^2 + w_A^{*2} \sigma^2(e_A) \quad S_P^2 = S_M^2 + \left[\frac{\alpha_A}{\sigma(e_A)} \right]^2$$

$$\text{Fund turnover} = \frac{\text{total value of the securities sold}}{\text{average value of the assets managed by the fund.}}$$

$$\text{Tax cost ratio} = \left[1 - \frac{(1 + TAR)}{(1 + PTR)} \right] \times 100,$$

$$t_i = \frac{E(R_i) - E(R_f)}{\beta_i} \quad S_p = \frac{E(R_p) - E(R_f)}{\sigma_p}$$

$$a_p = R_{p,t} - \left[R_{f,t} + \beta_p (R_{m,t} - R_{f,t}) \right] \quad M_{p.}^2 = r_p + S_p * (\sigma_M - \sigma_p)$$

$$IR_p = \frac{\alpha_p}{\sigma(e_p)} \quad ST_i = \frac{E(R_i) - \tau}{DR_i} \quad = \sqrt{\frac{1}{n} \sum_{R < E(R)} (R_{it} - E(R))^2}$$

$$MDD = \frac{(Trough\ Value - Peak\ Value)}{Peak\ Value} \quad \beta_F = \frac{\sigma_p}{\sigma_b} \quad Calmar_p = \frac{r_p}{MDD}$$

$$\text{Net Selectivity} = \underbrace{(R_p - \beta(R_b - R_f) - R_f)}_{\text{Selectivity}} - \overbrace{(\beta_F - \beta)(R_b - R_f)}^{\text{Diversification}}$$

$$= \alpha - d$$

